

Assignment: Solid-State and Devices Physics (PHYS2213)

Physical constants (unless stated otherwise, $T = 300$ K):

$$q = 1.602 \times 10^{-19} \text{ C}, \quad k_B = 1.380649 \times 10^{-23} \text{ J/K}, \quad V_T = \frac{k_B T}{q} \approx 0.0259 \text{ V}, \quad h = 6.626 \times 10^{-34} \text{ J s},$$

ASSIGNMENT

Numerical Problems

P1. Particle in a 1D infinite well (quantized energies).

Given: Electron in an infinite well of width $L = 5$ nm.

Find: E_1, E_2, E_3 and the ground-state de Broglie wavelength.

P2. Tunneling through a rectangular barrier (WKB estimate).

Given: Barrier height $V_0 = 0.50$ eV, width $a = 2$ nm; electron energy $E = 0.10$ eV.

Find: Transmission probability T using $T \approx \exp(-2\kappa a)$, where $\kappa = \sqrt{2m_e(V_0 - E)}/\hbar$.

P3. FermiDirac vs Boltzmann occupancy & error.

Given: $T = 300$ K. Evaluate $f(E)$ at (a) $E - E_F = 0.20$ eV and (b) $E - E_F = 0.05$ eV using both exact FD and Boltzmann.

P4 Crystallography: Miller indices and angular relationships between planes

Given: Consider a cubic crystal system with:

- Lattice parameter: $a = 0.30 \text{ nm}$
- A specific crystal plane that intersects the three coordinate axes at the following positions:
 - x-axis intercept: a (one lattice parameter)
 - y-axis intercept: a (one lattice parameter)
 - z-axis intercept: $a/2$ (half a lattice parameter)
- The crystal structure is simple cubic (90 angles between all axes)

Find:

- (i) Determine the Miller indices (hkl) that characterize this crystal plane using the standard crystallographic procedure
- (ii) Calculate the angle θ between the normal vector to this plane and the crystallographic direction $[001]$ (which is parallel to the positive z-axis)

Note: Show all steps in the Miller index determination, including taking reciprocals and clearing fractions. For the angle calculation, use vector dot product methods and express your final answer in degrees.

P5 Band structure analysis: Effective mass determination from experimental dispersion data

Given: Experimental band structure measurement on a semiconductor crystal showing:

- The conduction band has a parabolic dispersion relation near its minimum: $E(k) = E_c + \frac{\hbar^2 k^2}{2m^*}$
- The band minimum occurs at $k = 0$ (direct gap semiconductor)
- At a specific momentum value $k = 0.7 \times 10^9 \text{ m}^{-1}$, the measured energy above the conduction band edge is: $E(k) - E_c = 0.050 \text{ eV}$
- The measurement is performed at low temperature to minimize thermal broadening effects

Find: Determine the effective mass ratio m^*/m_e , where:

- m^* is the effective mass of electrons in this conduction band
- $m_e = 9.11 \times 10^{-31}$ kg is the free electron mass

Instructions: Use the parabolic band approximation and show all steps in your calculation. Discuss whether this effective mass value suggests the material would have high or low electron mobility compared to free electrons.

Questions

Diffusion and the Einstein Relation in intrinsic semiconductors

The transport of charge carriers in semiconductors involves both drift (due to electric fields) and diffusion (due to concentration gradients). The diffusion current density is given by:

$$J_{\text{diff}} = -qD \frac{dn}{dx}$$

At thermal equilibrium, drift and diffusion currents must balance.

- Starting from the principle of detailed balance at equilibrium, derive the Einstein relation between the diffusion coefficient D and the mobility μ .
- For a given semiconductor material, discuss how the diffusion coefficient depends on temperature and explain the physical mechanisms behind this temperature dependence.

Carrier Recombination

Carrier recombination is a fundamental process in semiconductors where electrons and holes combine, annihilating each other and reducing the excess carrier population. Different recombination mechanisms dominate under different material conditions and operating regimes. Understanding when each mechanism is dominant is crucial for device design and optimization.

- Describe the physical process of radiative recombination and explain why this mechanism is particularly efficient in direct bandgap semiconductors (such as GaAs, InP) compared to indirect bandgap semiconductors (such as Si, Ge).
- Explain the multi-step process of Shockley-Read-Hall (Trap-Assisted) recombination: (1) electron capture by a trap, (2) hole capture by the same trap, and (3) the subsequent annihilation. Why does this process occur more readily through trap states than through direct band-to-band recombination in defective materials?

(c) Analyze the temperature dependence of Auger recombination.